23-2g “. . . In a Given Period of Time.”

GDP measures the value of production that takes place within a specific interval of time. Usually, that interval is a year or a quarter (three months). GDP measures the economy’s flow of income, as well as its flow of expenditure, during that interval.

When the government reports the GDP for a quarter, it usually presents GDP “at an annual rate.” This means that the figure reported for quarterly GDP is the amount of income and expenditure during the quarter multiplied by four. The government uses this convention so that quarterly and annual figures on GDP can be compared more easily.

In addition, when the government reports quarterly GDP, it presents the data after they have been modified by a statistical procedure called *seasonal adjustment*. The unadjusted data show that the economy produces more goods and services during some times of the year than during others. (As you might guess, December’s holiday shopping season is a high point.) When monitoring the economy, economists and policymakers often want to look beyond these regular seasonal changes. Therefore, government statisticians adjust the quarterly data to take out the seasonal cycle. The GDP data reported in the news are always seasonally adjusted.

Now let’s repeat the definition of GDP:

- Gross domestic product (GDP) is the market value of all final goods and services produced within a country in a given period of time.

This definition focuses on GDP as total expenditure in the economy. But recall that every dollar spent by a buyer of a good or service becomes a dollar of income to the seller of that good or service. Therefore, in addition to adding up total expenditure in the economy to calculate GDP, the government also adds up total income in the economy to arrive at *gross domestic income* (GDI). GDP and GDI give almost exactly the same number. (Why “almost”? The two measures should be precisely the same, but data sources are not perfect. The difference between GDP and GDI is called the *statistical discrepancy*.)

It should be apparent that GDP is a sophisticated measure of the value of economic activity. In advanced courses in macroeconomics, you will learn more about the nuances of its calculation. But even now you can see that each phrase in this definition is packed with meaning.

FYI

Other Measures of Income

When the U.S. Department of Commerce computes the nation's GDP, it also computes various other measures of income to get a more complete picture of what's happening in the economy. These other measures differ from GDP by excluding or including certain categories of income. What follows is a brief description of five of these income measures, ordered from largest to smallest.

- *Gross national product* (GNP) is the total income earned by a nation's permanent residents (called *nationals*). It differs from GDP in that it includes income that our citizens earn abroad and excludes income that foreigners earn here. For example, when a Canadian citizen works temporarily in the United States, her production is part of U.S. GDP, but it is not part of U.S. GNP. (It is part of Canada's GNP.) For most countries, including the United States, domestic residents are responsible for most domestic production, so GDP and GNP are quite close.

- *Net national product* (NNP) is the total income of a nation's residents (GNP) minus losses from depreciation. *Depreciation* is the wear and tear on the economy's stock of equipment and structures, such as trucks rusting and old computer models becoming obsolete. In the national income accounts prepared by the Department of Commerce, depreciation is called the "consumption of fixed capital."

- *National income* is the total income earned by a nation's residents in the production of goods and services. It is almost identical to net national product. These two measures differ because of the *statistical discrepancy* that arises from problems in data collection.

- *Personal income* is the income that households and noncorporate businesses receive. Unlike national income, it excludes *retained earnings*, the income that corporations earn but do not pay out to their owners. It also subtracts indirect business taxes (such as sales taxes), corporate income taxes, and contributions for social insurance (mostly Social Security taxes). In addition, personal income includes the interest income that households receive from their holdings of government debt and the income that households receive from government transfer programs, such as welfare and Social Security.

- *Disposable personal income* is the income that households and noncorporate businesses have left after satisfying all their obligations to the government. It equals personal income minus personal taxes and certain nontax payments (such as traffic tickets).

Although the various measures of income differ in detail, they almost always tell the same story about economic conditions. When GDP grows rapidly, these other measures of income tend to grow rapidly. And when GDP falls, these other measures tend to fall as well. As a result, for monitoring fluctuations in the overall economy, it does not matter much which measure of income we use.

QuickQuiz

3. If the price of a hot dog is $\$2$ and the price of a hamburger is $\$4$, then 30 hot dogs contribute as much to GDP as ____ hamburgers.

    a. 5

    b. 15

    c. 30

    d. 60

4. Angus the sheep farmer sells wool to Barnaby the knitter for $\$20$. Barnaby makes two sweaters, each of which has a market price of $\$40$. Collette buys one of them, while the other remains on the shelf of Barnaby's store to be sold later. What is GDP here?

    a. $\$40$

    b. $\$60$

    c. $\$80$

    d. $\$100$

5. After graduation, an American college student moves to Japan to teach English. Her salary is included

    a. only in U.S. GDP.

    b. only in Japan's GDP.

    c. in both U.S. GDP and Japan's GDP.

    d. in neither U.S. GDP nor Japan's GDP.